Every term begins with either a variable or a left parenthesis

The Open Logic Project contributors

openlogic:lambda-calculus/syntax:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-lambda-calculus-syntax/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the full three-rule case text is verbatim in the span; every rule cited by label exists in the frozen syntax-unit section named in the source_reference.

Editorial note. The label lem:term-start names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Formation rules of terms. Syntax unit, definition \ollabel{defn:term} with rules \ollabel{defn:term-var}, \ollabel{defn:term-abs}, \ollabel{defn:term-app}. Case analysis of the proof: the variable rule produces a variable, the abstraction and application rules produce strings opening with a parenthesis.

Source reference. Original definition defn-term: sources/olp-lambda-calculus-syntax/pinned-original-source.tex, source lines 1489–1489. Prerequisite pointer invoked at proof line(s) 1547; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-term-abs: sources/olp-lambda-calculus-syntax/pinned-original-source.tex, source lines 1494–1494. Prerequisite pointer invoked at proof line(s) 1549; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-term-app: sources/olp-lambda-calculus-syntax/pinned-original-source.tex, source lines 1496–1496. Prerequisite pointer invoked at proof line(s) 1549; it names a standard fact of the same unit that

the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition defn-term-var: sources/olp-lambda-calculus-syntax/pinned-original-source.tex, source lines 1492–1492. Prerequisite pointer invoked at proof line(s) 1548; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *A term starts with either a variable or a parenthesis.*

Proof. Something counts as a term only if it is constructed according to defn-term. If it is the result of defn-term-var, it must be a variable. If it is the result of defn-term-abs or defn-term-app, it starts with a parenthesis. □